

Same Counts, Different Message

AP Statistics · Unit 1 · Topic 1.4 · TEACHER KEY

CED TETHER

- Enduring Understanding: UNC-1 Graphical representations and statistics allow us to identify and represent key features of data.
- Skills:
- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- Learning Objectives and Essential Knowledge:
- LO UNC-1.C Represent categorical data graphically. [Skill 2.B]
- EK UNC-1.C.1 Bar charts (or bar graphs) are used to display frequencies (counts) or relative frequencies (proportions) for categorical data.
- EK UNC-1.C.2 The height or length of each bar in a bar graph corresponds to either the number or proportion of observations falling within each category.
- EK UNC-1.C.3 There are many additional ways to represent frequencies (counts) or relative frequencies (proportions) for categorical data.
- LO UNC-1.D Describe categorical data represented graphically. [Skill 2.A]
- EK UNC-1.D.1 Graphical representations of a categorical variable reveal information that can be used to justify claims about the data in context.
- LO UNC-1.E Compare multiple sets of categorical data. [Skill 2.D]
- EK UNC-1.E.1 Frequency tables, bar graphs, or other representations can be used to compare two or more data sets in terms of the same categorical variable.

Essential question: How can a graph use correct counts but still create a misleading comparison?

DOK-3 skill: 4.B · **FRQ pattern:** truncated-axis-which-display-is-honest-and-what-it-misleads

DOK rationale: Part (c) judges two headlines by relating the graph baseline to visible bar heights and contrasting that impression with the actual counts.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- What changed between the graphs if the votes are identical?
- Where does each visible bar begin?
- What do the counts support about the size of Pizza's lead?
- What headline would tell readers both the leader and the size of the difference?

[WARN] LOOK FOR

- Calling Graph A's counts false instead of explaining its baseline
- Treating the visible heights as vote counts

- Repeating a headline without using the counts

SCORING PART (c) — E / P / I

Expected elements

- **graph-and-scale** (required) — Chooses B and explains zero baseline makes heights proportional to counts
- **compares-impression** (required) — Contrasts Graph A's 12-to-1 visible heights with 52 versus 41 or an 11-vote difference
- **judges-headlines** (required) — Judges both headlines and supplies a bounded headline supported by the counts

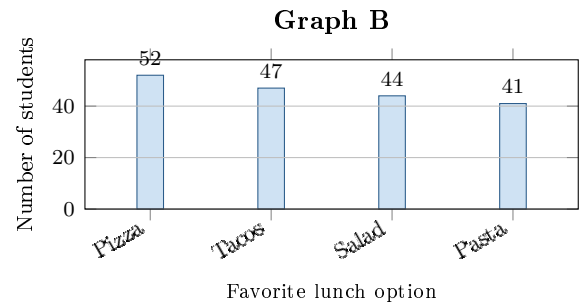
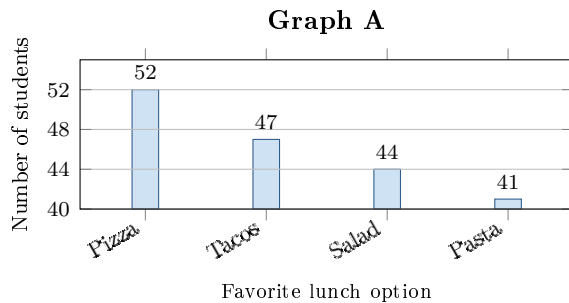
E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

- Calling Graph A's counts false instead of explaining its baseline
- Treating the visible heights as vote counts
- Repeating a headline without using the counts

[STAR] THE PROBLEM — annotated

Suppose 184 students chose a favorite lunch option. A newsletter printed **Graph A** with the headline “**Pizza wins by a landslide.**” A council member redrew the data as **Graph B** and said, “It is basically a tie.”



FIRST TAKE — one sentence before you work the parts

Is pizza clearly the favorite? Name the graph that most influenced you.

Key: Not graded. Expect Graph A to attract votes because its Pizza bar dominates.

[BOOK] GRAPHING ONE CATEGORICAL VARIABLE (keep on the page)

Bar height represents a category’s count or relative frequency. Label both axes, use equal-width bars, and start the vertical scale at zero to avoid distortion. The visible height above a displayed baseline is the count minus that baseline. Compare what the lengths suggest with the actual counts before writing a claim.

DOK 1 (a) Read the Pizza and Pasta counts. How many more students chose Pizza?

Key: Pizza has 52 students and Pasta 41, so Pizza has 11 more.

DOK 2 (b) Graph A starts at 40. Find the visible heights above 40 for Pizza and Pasta. Compare those heights with the actual counts.

Key: Visible heights: Pizza $52 - 40 = 12$ and Pasta $41 - 40 = 1$. Pizza looks 12 times as tall, while the actual counts are 52 and 41, a difference of 11 students.

DOK 3 (c) Which graph gives a fairer picture of the lunch votes? Explain how its scale affects the comparison. Use your counts to judge both ‘landslide’ and ‘basically a tie,’ then write a more accurate headline.

Key: Graph B uses zero, so full bar heights represent counts. Graph A starts at 40 and exaggerates the difference: visible heights are 12 to 1, but counts are 52 to 41. Pizza has the most votes, yet neither a huge landslide nor an exact tie describes the counts. A supported headline is ‘Pizza leads Pasta by 11 votes; all four options receive support.’ Accept other measured wording supported by the counts.